

Is having a large population beneficial for a country?

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1. Introduction

Population economics (or demographic economics) studies the relationship between populations and economic situations. The field has remained mostly within the straitjacket of short-run analysis, which boils down to the deduction that more people sharing the same output, or the same land and tools, implies less output per person (Simon, 1992). We might thus expect that a larger population results in lower life quality, potentially resulting in measurable declines in key indicators like GDP per capita and life expectancy.

Could population size have a negative effect on life expectancy and GDP per capita?

- **Prediction 1:** Countries with larger populations tend to have lower life expectancy
- **Prediction 2:** Countries with larger populations tend to have lower GDP per capita

2. Data

The dataset used comes from Moodle (accessed on 19.11.2025). The dataset contains various information about several countries including macroeconomic and demographic indicators like their population, life expectancy and GDP per capita. The data correspond to the year 2015 ($N = 119$, see **Figure 1**).

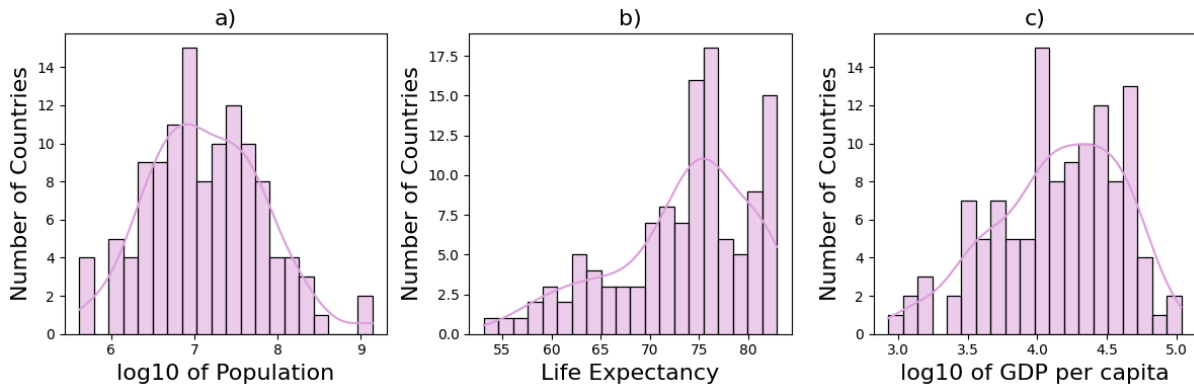


Figure 1 - Monovariate descriptions of the dataset: (a) log10 of Population, (b) Life Expectancy, (c) log10 of GDP per capita of countries.

3. Methods

In order to test the two predictions, we performed two linear regressions comparing (1) life expectancy depending on log10 of population, and (2) log10 of GDP per capita depending on log10 of population.

4. Results

Countries with a larger population have a lower life expectancy ($b = -2.22$, $R^2 = 0.047$, $p = 0.018$, see **Figure 2.a.**). Countries with a larger population have lower GDP per capita ($b = -0.18$, $R^2 = 0.077$, $p = 0.002$, see **Figure 2.b.**).

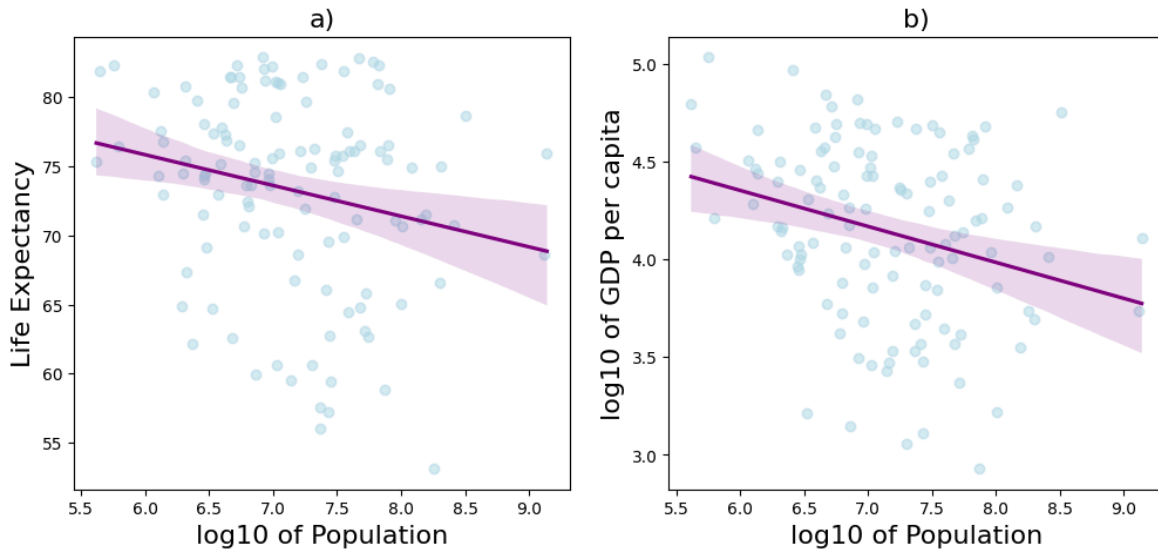


Figure 2 - Bivariate description of the dataset: (a) Life expectancy depending on log10 of population, and (b) log10 of GDP per capita depending on log10 of population.

5. Conclusion

Our results indicate that countries with larger populations tend to have a lower life expectancy and lower GDP per capita. These findings suggest that larger populations may pose challenges to the well-being of countries.

It should be noted that this study is not causal. Population size may correlate with other unobserved factors that influence both results.

6. References

SIMON, J. L. (1992). *Population and Development in Poor Countries: Selected Essays*. Princeton University Press. JSTOR. <https://doi.org/10.2307/j.ctt7zvf61>.